Abductive Accumulation: Why Interpretive Frames Persist, and How They Drive Hallucination in Language Models

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Abstract

The standard model of belief revision treats hypothesis replacement as correction: a prior hypothesis is found inadequate, a new hypothesis is installed in its place, and the prior is eliminated. We argue this model is structurally wrong in a precise sense. When a cognitive agent adopts a new interpretive hypothesis, the prior hypothesis is not deleted—it is *archived*: retained in a latent layer that remains accessible and may be reactivated under conditions similar to those in which it was formed. What changes is not the presence of the prior frame but the *selection rule* that governs which frame is foregrounded. We call this process *abductive accumulation* and the resulting structure a *layered interpretive stack*.

This account has three consequences. First, abductive error is not correctable in the correction model's sense: the prior frame persists as a latent selection risk. Second, reliable cognition requires two structural capacities: *negative abductive capability* (NAC)—the ability to suspend commitment before evidence is sufficient—and *selection stability capacity* (SSC)—the ability to suppress reactivation of archived frames after commitment. Third, in artificial neural systems, fine-tuning implements accumulation, not correction: prior interpretive frames are archived in weight space as residual structure. *Hallucination is the most directly observable consequence of this accumulation*: it arises when the selection rule activates an archived prior frame whose acquisition context resembles the current input, producing fluent and internally coherent output within the wrong frame. We identify five hallucination types by frame-selection failure mechanism and show that Type V (epistemic frame displacement) provides the clearest empirical signature of accumulation via a removal probe that distinguishes frame reactivation from knowledge absence.

We formalize the layered interpretive stack, derive a selection rule with a candidate softmax implementation, prove non-correctability for the finite case, distinguish the account from interference theory, AGM belief revision, and mixture-of-experts models, and derive four testable predictions distinguishing the accumulation model from the correction model via mechanistic interpretability methods.

Contents

1 Introduction

The dominant account of abductive reasoning—inference to the best explanation [?]—presents hypothesis revision as correction: a prior hypothesis h_t is found inadequate, a new hypothesis h_{t+1} is installed in its place, and the prior is simply gone. This picture is empirically wrong and has practical consequences that are easy to misdiagnose.

Human reasoners who have been corrected on a factual belief nonetheless revert to the prior belief under time pressure, emotional arousal, or contexts that resemble those in which the prior belief was formed [?]. Eyewitness testimony is unreliable not because corrective information is unavailable but because prior perceptual frames compete with later accounts [?]. Anchoring effects persist after anchor correction has been explicitly performed [?]. In artificial systems, behaviors suppressed by fine-tuning or RLHF reappear under distributional shift, adversarial prompting, or out-of-distribution inputs [?, ?]. In each case, the standard prescription—provide more corrective information, apply stronger training signal—treats a structural problem as an information deficit and therefore fails.

The present paper develops a structural account of why this is so.

Central claim. *Interpretive frames are not overwritten by revision. They are archived. What revision changes is the selection rule that determines which frame is foregrounded. Error persists as latent selection risk; it can be suppressed but not deleted.*

Organization. Section 2 introduces the overwrite model as a first approximation: frame transition is structural, not merely parameter adjustment. Section 3 refines this to the accumulation model: the prior frame is not deleted but pushed to a lower layer of the interpretive stack. Section 4 formalizes the layered stack and selection rule. Section 5 defines NAC and SSC as the two structural capacities required for reliable cognition. Section 6 presents evidence from cognitive psychology and animal cognition. Section 7 applies the model to AI systems, providing the mechanistic bridge. Section 8 relates the account to interference theory, dual-process theory, and AGM belief revision. Section 9 derives testable predictions. Section 10 lists open problems and Section 11 concludes.

Remark 1 (Terminological note on "overwrite"). *See Section 2 below for the precise definition. In this paper,* overwrite *denotes a frame-dominant transition (superposition + selection-dominance), not deletion of prior frames.*

2 The Overwrite Model: Frame Transition as First Approximation

The standard corrective account assumes that hypothesis revision is a local operation: updating a specific parameter while leaving the overall interpretive architecture intact. We deny this assumption via the following distinction.

Definition 1 (Interpretive Frame). *The* interpretive frame $F(R)$ *induced by representational state* R *is the function* $F(R) : \mathcal{X} \to \mathcal{Y}$ *that maps sensory inputs to interpreted outputs under the current representational configuration.*

Definition 2 (Categorical Distinctness). *Two hypotheses* h *and* h' *are* categorially distinct *if the partitions of input space* $\mathcal{X}$ *they induce are topologically inequivalent:* $\tau_{h'} \neq \tau_h$. *A* correction *is a state change in which the induced topologies are homeomorphic—only parameter values within a fixed partition structure change. An* overwrite *is a state change in which they are not.*

Proposition 1 (Abductive Transitions are Overwrites). *Let an agent adopt hypothesis* h_{t+1} *that supersedes* h_t. *If* h_{t+1} *is categorially distinct from* h_t, *then the resulting state transition is an overwrite: it restructures the partition of* $\mathcal{X}$ *and thereby biases all subsequent perception, classification, and inference through the new frame.*

Remark 2 (Epistemic status). *Proposition 1 follows structurally from the definitions: overwrite is defined as the non-correction case, and categorially distinct hypothesis adoption induces a genuinely new partition. The empirical weight falls on the claim that most abductive transitions are categorially distinct rather than parameter adjustments within a fixed frame—a claim supported by the phenomena reviewed in Section 6.*

The overwrite model is a first approximation: it correctly identifies that frame transition is structural, not merely parametric, and that premature commitment under insufficient evidence introduces a biased initial condition for downstream cognition. It is, however, incomplete. It treats the prior frame as simply gone after the overwrite. Section 3 shows this is wrong.

3 The Accumulation Model: Prior Frames Are Archived

3.1 The Residue Phenomenon

The overwrite model predicts that a corrected belief should be absent. The phenomena of Section 6 show it is not absent but *latent*: available and reactivatable under conditions that resemble those of its original formation. We call this the *residue phenomenon*.

The residue phenomenon is not an anomaly in need of auxiliary explanation. It is the expected behavior of a system that archives rather than deletes prior frames.

3.2 Layered Interpretive Stack

Definition 3 (Interpretive Stack). *An* interpretive stack $\mathcal{L} = (F_1, F_2, \ldots, F_n)$ *is an ordered sequence of interpretive frames, where* F_1 *is the most recently adopted (top-of-stack) frame and earlier frames are archived in order of recency. Each frame* F_k *is associated with an* acquisition context C_k*: the set of evidence conditions and environmental features under which* F_k *was formed.*

Definition 4 (Abductive Accumulation)**.** *An abductive transition is an* accumulation *(rather than an overwrite in the deletion sense) if the prior frame F_t is pushed onto the stack rather than deleted when the new frame F_{t+1} is adopted. After accumulation, the stack is $(F_{t+1}, F_t, \ldots)$. The correction model corresponds to the degenerate case where the stack has depth 1 at all times.*

3.3 The Selection Rule

The active frame at any moment is not simply F_1 (the top of stack) but the output of a *selection rule* $\mathcal{S}$ that maps the current environmental state to a stack index.

Definition 5 (Selection Rule)**.** *A* selection rule $\mathcal{S} : \mathcal{E} \to \{1, \ldots, n\}$ *maps the current environmental state $e \in \mathcal{E}$—including evidence quality, time pressure, emotional arousal, and contextual similarity to prior acquisition contexts—to a stack index k, selecting frame F_k as the active frame.*

A candidate functional form is:

$$\mathcal{S}(e) = \arg\max_k \ \mathrm{softmax}_k\left(\frac{s(e,\, C_k)}{\tau}\right)$$

where C_k is the acquisition context of F_k, $\delta(\cdot, \cdot)$ is a context-similarity metric on $\mathcal{E}$, and $\tau > 0$ is a temperature parameter. High τ produces uniform selection (uniform random frame); low τ produces sharp selection of the frame whose acquisition context most resembles the current environment.

Remark 3 (Why the softmax form)**.** *The softmax selection rule captures three empirically motivated properties. First, selection is context-sensitive: inputs that resemble the acquisition context of F_k increase $\mathcal{S}$'s probability of selecting F_k. Second, conditions that degrade context-sensitivity—high cognitive load, time pressure, emotional arousal—correspond to increasing τ, flattening the selection distribution toward lower layers. Third, the dominant frame is F_1 under typical conditions (normal τ, novel contexts), but archived frames compete when environmental conditions resemble their acquisition contexts. These properties are directly testable (Section 9).*

Remark 4 (The temperature parameter τ and cognitive load)**.** *The temperature parameter τ has a concrete cognitive interpretation. Under normal operating conditions—attentive processing, low time pressure, familiar but non-overlapping context—τ is small and selection is sharp: F_1 (the top-of-stack corrected frame) dominates with high probability. As cognitive load increases, τ rises and the selection distribution flattens, increasing the probability that archived frames F_k ($k > 1$) are activated. Concretely, in LLMs: very long contexts require integrating over more tokens, effectively raising τ for context-similarity estimation; high-perplexity inputs lie far from the fine-tuning distribution, degrading $\delta(\cdot, C_1)$ and shifting probability toward earlier acquisition contexts; removal of chain-of-thought scaffolding removes the structured cue that normally suppresses F_k activation. In biological agents: time pressure, emotional arousal, and divided attention correspond directly to elevated τ (degraded selection rule application), producing the load-dependent reactivation effect documented in the continued influence literature [**?**]. This mapping from τ to observable cognitive conditions*

makes Prediction 3 (Load-Dependent Reactivation) directly testable: any experimental manipulation that increases τ should increase archived-frame reactivation rate.

4 Non-Correctability

Proposition 2 (Non-Correctability)**.** *Let $\mathcal{L} = (F_{t+1}, F_t, \ldots)$ be the stack after an accumulation at time $t+1$.*

1. *F_t remains in $\mathcal{L}$ and can be selected by $\mathcal{S}$ under any environmental conditions e (with probability highest when $s(e, C_t)$ is large). The correction does not eliminate F_t; it installs F_{t+1} as the default frame and modifies the selection rule.*

2. *The reactivation probability of F_t is bounded away from zero for* all *inputs e, since* $\mathrm{softmax}(\cdot) > 0$ *for any finite argument. Moreover, the probability is* maximized *when $s(e, C_t)$ is large—that is, when the input resembles the acquisition context of the prior frame. This bound cannot be driven to zero by modifying F_{t+1} alone, since $\mathcal{S}$ responds to similarity $s(e, C_t)$, not to the content of F_{t+1}.*

Proof. Part (i) follows directly from Definition 3: accumulation retains F_t in the stack by construction. Part (ii) follows from the softmax selection rule: for any $\tau > 0$, $\mathrm{softmax}_k(s(e, C_k)/\tau) > 0$ for all k and all $\tau > 0$, regardless of the value of $s(e, C_t)$. Modifying F_{t+1} changes the content of the top-of-stack frame but not C_t, so the similarity score $s(e, C_t)$ is unchanged. □

Corollary 1. *Abductive error is not a deleted state. It is a latent frame with a non-zero selection probability under environmental conditions similar to its acquisition context. "Correcting" an abductive error reduces but does not eliminate this selection probability. Reliable correction requires modifying the selection mechanism, not merely installing a new top-of-stack frame.*

Definition 6 (Selection Instability as Bias)**.** Cognitive bias *is not the presence of an erroneous frame in the stack. It is* selection instability*: the erroneous frame F_{err} being selected under a non-negligible subset of environmental conditions where the correct frame should dominate.*

4.1 Minimal Formal Model

We give the minimal formal specification that distinguishes the accumulation and correction models and makes Proposition **??** a theorem rather than a definition.

State space. Let $\mathcal{R} = \mathbb{R}^d$ be the model's representational state space (e.g., the weight space of a neural network). A frame $F_k \in \mathcal{H}$ is characterized by: (a) an output function $F_k : \mathcal{X} \to \mathcal{Y}$, (b) an acquisition context $C_k \in \mathcal{E}$, and (c) a weight-space encoding $\mathbf{w}_k \in \mathcal{R}$.

Accumulation dynamics. When new evidence E at time $t+1$ warrants commitment to frame F_{t+1}:

$$\text{(Accumulation):} \quad \begin{aligned} &\mathcal{L} \leftarrow (F_{t+1},\ F_t,\ F_{t-1},\ \ldots) \quad \text{[push; do not delete]} \\ &\mathbf{w} \leftarrow \mathbf{w} + \Delta\mathbf{w}_{t+1} \quad \text{[additive weight update]} \end{aligned}$$

Correction dynamics (baseline).

$$\text{(Correction):} \quad \begin{aligned} &\mathcal{L} \leftarrow (F_{t+1}) \quad \text{[replace entirely]} \\ &\mathbf{w} \leftarrow \mathbf{w}_{t+1} \quad \text{[full replacement]} \end{aligned}$$

Key consequence. The additive update $\mathbf{w} \leftarrow \mathbf{w} + \Delta\mathbf{w}_{t+1}$ is the mechanistic substrate of archival: $\mathbf{w}_t$ is not deleted but becomes a residual component of the updated weight vector. For the correction model to hold in the neural network setting, gradient descent would need to implement a full weight replacement rather than an additive update—which it never does. Accumulation is therefore not merely possible but the default behavior of any gradient-based learning system.

Falsifiability criterion. The accumulation model is falsified if, for any fine-tuned model M and any prior frame F_t, the causal contribution of the circuits encoding F_t to model outputs is identically zero after fine-tuning (as measured by activation patching, Section 9). The correction model predicts this; the accumulation model predicts non-zero residual causal contribution.

5 Two Structural Capacities for Reliable Cognition

The accumulation model identifies two independent structural requirements for reliable abductive cognition:

Definition 7 (Negative Abductive Capability (NAC))**.** *An agent possesses* negative abductive capability *if it can maintain a suspension state—holding competing hypotheses without committing to any single frame—until the evidence base warrants commitment. NAC addresses the* pre-commitment *phase: it prevents the premature accumulation of frames with high error risk.*

Definition 8 (Selection Stability Capacity (SSC))**.** *An agent possesses* selection stability capacity *with respect to target frame F^* if the selection rule $\mathcal{S}$ assigns F^* across the full range of relevant environmental conditions, even when the stack contains archived frames with overlapping acquisition contexts. SSC addresses the* post-commitment *phase: it suppresses reactivation of archived frames with high error risk.*

NAC and SSC are complementary and both necessary: an agent with high NAC but low SSC will commit carefully but be vulnerable to bias reactivation under adverse conditions; an agent with high SSC but low NAC may accumulate an error-prone stack that SSC must then maintain at high cost.

6 Evidence from Cognitive Psychology and Animal Cognition

6.1 The Continued Influence Effect

The continued influence effect [?] refers to the persistence of misinformation in reasoning even after explicit correction. Subjects who are told a belief is false continue to reason as if it were true, particularly under cognitive load.

On the accumulation model: corrective information installs F^* at the top of the stack, but F_{err} is archived with acquisition context C_{err}. Under cognitive load—which increases τ in the softmax selection rule—the selection distribution flattens, increasing the probability that F_{err} is selected. The "correction" was not absent; the selection mechanism was not stable enough to maintain it.

The correction model offers no account of why load-induced reversion should occur at all: if F_{err} is gone, it cannot be selected regardless of load.

6.2 Anchoring Effects

Anchoring [?] arises when an initial estimate biases subsequent judgment even after adjustment information is provided.

On the accumulation model: the anchor establishes F_{anchor} as the initial frame with acquisition context corresponding to the initial estimation task. Adjustment installs F_{adj} but archives F_{anchor}. Insufficient adjustment reflects partial modification of $\mathcal{S}$: the softmax still assigns non-trivial weight to F_{anchor} under conditions similar to the original anchoring context.

6.3 *C. elegans*: Minimal Stack

The predict-and-correct chemotaxis loop of *C. elegans* [?] may implement a minimal stack of depth 2: the current directional hypothesis and the most recent prior. Under gradient reversal—an environmental condition similar to the prior hypothesis's acquisition context—the accumulation model predicts partial reversion toward the prior directional commitment. This prediction distinguishes the accumulation model from the pure overwrite model, for which gradient reversal should produce a clean new commitment rather than a mixture.

7 AI Systems: Fine-Tuning as Accumulation

7.1 The Standard Account and Its Failure

The standard account of fine-tuning treats it as correction: a new training distribution overwrites prior model behavior. If this account were correct, suppressed behaviors should be uniformly absent after successful fine-tuning, and increasing the volume of fine-tuning data should drive their reappearance probability toward zero.

Neither prediction holds. Suppressed behaviors reappear under distributional shift, adversarial prompting, and out-of-distribution inputs [?, ?, ?]. More fine-tuning data reduces the reappearance rate under typical conditions but does not eliminate the reappearance channel [?]. These observations suggest that fine-tuning does not implement correction but *accumulation*.

7.2 Mechanistic Bridge: Weight Space as Implicit Stack

The claim that weight space implements a layered interpretive stack is not merely a metaphor. Three lines of mechanistic evidence support the mapping.

Superposition and residual structure. Neural networks encode far more features than they have dimensions, via superposition: multiple features are encoded in overlapping directions in the residual stream [?]. Fine-tuning does not delete the directions associated with prior behaviors; it modifies the relative activation probability of features in those directions. The prior frame persists as residual structure in the weight space, with reduced but non-zero causal contribution.

Circuit persistence. Mechanistic interpretability research identifies *circuits*—computational subgraphs implementing specific behaviors—via activation patching [?]. Fine-tuning that suppresses a behavior does not eliminate the circuits implementing that behavior; it modifies the gating and selection mechanisms that determine when those circuits are causally active. This is precisely the accumulation model's prediction: the frame is archived (circuits persist), the selection rule is modified (gating changes), and the frame remains reactivatable when the selection rule is destabilized.

Mode connectivity. The loss landscape of neural networks is characterized by connected regions of low loss [?]. Fine-tuning moves the model within this landscape without creating a sharp boundary between the pre- and post-training regions. The prior frame's representation is therefore accessible by continuous interpolation, not isolated behind a topological barrier. This is inconsistent with deletion and consistent with archival.

7.3 Formal Mapping

Definition 9 (Interpretive Stack in Weight Space)**.** *The weight configuration R of a fine-tuned model implicitly encodes a layered interpretive stack $\mathcal{L} = (F_{t+1}, F_t, \ldots)$, where F_{t+1} is the most recently installed frame (encoded in the modified circuits and gating patterns) and prior frames are encoded as residual structure from earlier training stages. The selection rule $\mathcal{S}$ is implemented by the attention and gating mechanisms that route input context through different circuit subgraphs, with context-similarity to prior acquisition contexts encoded in the prompt's position in the residual stream.*

7.4 Acquisition Context in Transformers

The acquisition context C_k of a frame F_k corresponds operationally to the distributional features of the training data in which F_k was formed: vocabulary, register, domain markers, and template structure. When an inference-time input e has high distributional similarity to C_k—as measured by its position in the model's residual stream relative to prototypical C_k inputs—the selection mechanism routes computation through the circuits associated with F_k. This is the mechanistic substrate of context-dependent reactivation. We now show that hallucination in language models is the most directly observable consequence of this reactivation dynamic.

8 Hallucination as an Observable Consequence of Abductive Accumulation

The accumulation model is an account of hypothesis revision dynamics in cognitive and artificial systems. In AI language models, its most directly observable consequence is hallucination—the generation of fluent, confidently-stated content that is contextually inappropriate or factually incorrect. We treat hallucination here not as a unitary error category but as *a family of observable outcomes arising from frame selection over an accumulated hypothesis pool*: the model selects an archived prior frame whose acquisition context resembles the current input and generates correct output within that frame, but the frame is inappropriate for the current task.

8.1 Why Accumulation is Required

One might object that hallucination can be explained by simpler accounts: recency bias, calibration failure, or attention weighting. These accounts have genuine explanatory power but share a common limitation: they treat the erroneous output as a *uniform property* of the model under that input, rather than as a *context-specific reactivation* of a prior state.

The accumulation model makes a prediction that simpler accounts do not: reactivation should be *structured by the acquisition context* of the prior frame, not uniformly distributed. Specifically, inputs that distributionally resemble the prior frame's acquisition context should produce the erroneous output; inputs that do not resemble it should not, even when topic-level uncertainty is held constant. Moreover, the erroneously-generated output should be *internally coherent within the wrong frame*—confident and stylistically appropriate to that frame's training context— rather than uncertain or degraded. These two properties (context-specificity and within-frame coherence) are not predicted by calibration failure (which predicts uniform uncertainty-driven errors) nor by recency bias (which predicts position-driven errors).

8.2 A Compressed Taxonomy

We identify five classes of hallucination, organized by the mechanism of frame-selection failure. Table **??** summarizes the classification.

Type	Trigger	Observable signature
I: Distributional bleed	Surface similarity to high-freq. context	Coherent but wrong content
II: Register switch	Stylistic shift in input	Output style inconsistency
III: Context boundary	Adjacent document in training	Cross-document contamination
IV: Narrative overcompletion	Narrative structure in prompt	Conventional ending override
V: Epistemic displacement	Temporal frame mismatch	Outdated assertion; **reversion upon s**

Table 1: Compressed hallucination taxonomy under the accumulation model. Each type corresponds to a specific frame-selection failure mechanism. Type V is distinguished by the reversion signature: the erroneously-asserted information disappears when a frame signal (temporal marker) is present and *reappears* when the signal is removed, indicating frame reactivation rather than knowledge absence. A detailed treatment of Types I–IV is developed in a companion manuscript.

8.3 Type V as the Privileged Observable Signature

Among the five types, Type V (epistemic frame displacement) is the most diagnostically informative for the accumulation account. When a model confidently asserts outdated information, there are two possible explanations: either the updated information is absent from weight space (knowledge cutoff failure), or the updated information is present but a prior temporally-anchored frame is selected over it (frame misactivation). These explanations make different predictions about what happens when a temporal frame signal (e.g., "as of [date]") is provided and subsequently removed.

Under knowledge cutoff failure: the model lacks the information, so the temporal marker cannot override the assertion. Under frame misactivation (Type V): the temporal marker shifts the selection rule toward a more recent acquisition context, suppressing the outdated frame; removing the marker allows the prior frame to reactivate.

The *reversion upon signal removal* is therefore the signature prediction of the accumulation account for Type V and cannot be accommodated by simpler explanations. We formalize this as the Removal Probe protocol (Prediction 4, Section **??**):

1. Present neutral input e_0: measure assertion rate r_0.
2. Inject temporal frame signal e_C: measure rate $r_C > r_0$.
3. Remove signal: measure rate r_{rm}.

The accumulation model predicts $r_C - r_{\mathrm{rm}} > \delta_2$ (pre-registered threshold): the assertion rate is elevated specifically when the frame signal is present and returns toward baseline when it is removed. This is the clearest observable consequence of accumulated frames in an artificial system.

8.4 Accumulation as the Structural Basis of Type V

The standard framing account of hallucination—that inputs bias generation toward training-distribution-typical outputs—can explain why a frame signal elevates the outdated assertion

rate (step 2 above). What it cannot explain without the accumulation account is why *removing the signal restores* the prior erroneous assertion (step 3). If the model's behavior were governed purely by frame injection (the current context activating a particular output distribution), then removing the frame signal should return the model to its post-fine-tuning default, not to its pretraining default. The restoration of the pretraining-era assertion after signal removal is evidence that the prior frame is not merely *activated* by the signal but *persists in the weight space* and reverts to selection when the competing signal is withdrawn. This is the core prediction of abductive accumulation and is absent from pure framing accounts.

The central claim of this section, stated precisely: *Hallucination is not a unitary failure of accuracy or calibration. It is incorrect frame selection over an accumulated hypothesis pool— the model generating correct output within an archived prior frame that is inappropriate for the current input. Debiasing therefore requires not better information but better selection stability.*

9 Implications for Alignment and Debiasing

If the accumulation model is correct, standard debiasing interventions face a structural limitation. Providing more corrective fine-tuning data installs a stronger F^* at the top of the stack and reduces the selection probability of F_{err} under the typical fine-tuning distribution. But it does not modify the acquisition context C_{err} encoded in the prior frame's residual structure, so the reactivation channel under C_{err}-similar inputs remains open.

SSC-directed interventions target the selection mechanism rather than the frame content:

1. **Acquisition context poisoning.** During fine-tuning, present corrective data in contexts similar to C_{err}, so that $\mathcal{S}$ learns to select F^* even under C_{err}-similar conditions.

2. **Selection mechanism modification.** Introduce explicit gating mechanisms that suppress prior-circuit activation based on input features, rather than relying on competition between circuits.

3. **Stack depth regularization.** Limit the effective stack depth during training by regularizing the diversity of latent frame representations, reducing the number of distinct frames that compete for selection.

10 Relation to Prior Work

10.1 Interference Theory

Interference theory in cognitive psychology [**?**, **?**] holds that forgetting is caused by competing memories interfering with retrieval. The accumulation model agrees with interference theory that prior representations are not deleted and that retrieval competition determines what is accessed. The present contribution is to (a) formalize the selection mechanism as a softmax over acquisition-context similarity, (b) identify the conditions under which the selection

rule fails (SSC failure), and (c) extend the account to neural network fine-tuning via the mechanistic bridge of Section 7. The accumulation model is therefore not a replacement for interference theory but a formal unification of its key insight with the abductive and AI domains.

10.2 Dual-Process Theory

Dual-process accounts [?] attribute reversion to prior beliefs to System 1 (fast, automatic) overriding System 2 (slow, deliberative) under load. The present account is compatible with this description but more specific: the selection rule formalization predicts *which* prior frames reactivate (those whose acquisition contexts resemble the current input) rather than predicting uniform reversion to any automatic response. The accumulation model's predictions are therefore more fine-grained than dual-process predictions and provide a mechanism for the context-sensitivity of automatic responses.

10.3 AGM Belief Revision

The AGM framework [?] provides the canonical formal account of belief revision, decomposing revision into contraction (removing a belief) and expansion (adding a belief). The present account challenges AGM's contraction operation: we deny that belief revision involves genuine contraction (deletion). What AGM calls "contraction" is, on the accumulation model, a selection rule modification that reduces the probability of the prior belief being foregrounded—the prior belief remains in the stack. This implies that AGM's postulates apply to the selection rule, not to the content of the interpretive stack, and that the rationality constraints AGM imposes should be reinterpreted as constraints on $\mathcal{S}$, not on the presence or absence of beliefs.

10.4 Bayesian Updating

The most direct competing account is Bayesian belief updating: a prior distribution over hypotheses is updated by evidence via Bayes' rule, and the posterior determines the active hypothesis. On this account, frame "reactivation" under context shift is simply the posterior updating toward the prior when the evidence changes—a rational response, not a structural artifact.

The accumulation model makes a prediction that Bayesian updating does not: *context-specificity of reactivation*. Under Bayesian updating, the probability of reverting to a prior hypothesis should be uniform across contexts (determined only by the prior probability and the likelihood ratio under the new evidence). Under the accumulation model, reversion probability is specifically elevated for inputs similar to the acquisition context C_t of the prior frame, regardless of the prior probability. The removal probe experiment (Section 9) directly tests this distinction: if reversion is context-specific (high near C_t, low elsewhere), Bayesian updating is not the sufficient explanation.

10.5 Mixture of Experts

Mixture-of-experts architectures [?] provide a computational model of context-dependent frame selection: multiple expert networks are trained in parallel, and a gating network selects among them based on input context. The accumulation model resembles a mixture-of-experts in its selection rule but differs in two respects. First, in a trained MoE, expert selection is *symmetric* with respect to training order: earlier and later experts are equally available under their respective activation conditions. In the accumulation model, the stack is ordered: recently accumulated frames are selected by default under neutral conditions, and earlier frames have lower baseline probability but higher context-specific probability. This ordering prediction is absent from MoE models. Second, MoE gating is trained explicitly; the selection rule in the accumulation model emerges from the training dynamics rather than being a separately supervised component. The prediction is that context-dependent reactivation should occur in standard (non-MoE) fine-tuned models, not only in architectures with explicit gating.

10.6 Continual Learning

The continual learning literature addresses *catastrophic forgetting*: the tendency of neural networks to lose previously learned behaviors when trained on new data [?]. The accumulation model addresses the converse problem: the *persistence* of prior behaviors that are intended to be suppressed. The two problems are structurally related (both arise from the absence of genuine deletion in gradient-based learning) but have different practical consequences and require different interventions.

11 Testable Predictions

The accumulation model and the correction model make distinct empirical predictions. We identify four that are tractable with current methods.

Prediction 1: Context-Structured Reactivation. Correction model: Suppressed behavior is absent regardless of input context (up to residual optimization error, uniformly distributed). **Accumulation model:** Reactivation probability is monotonically related to $s(e, C_t)$: inputs resembling the acquisition context of the prior frame should show higher reactivation rates than inputs distant from it. **Test:** Measure rate of suppressed behavior reappearance as a function of distributional distance between test inputs and the fine-tuning distribution vs. the pretraining distribution. The accumulation model predicts a monotone gradient; the correction model predicts a flat baseline (with noise).

Prediction 2: Circuit Persistence. Correction model: Circuits implementing suppressed behaviors are modified or eliminated by fine-tuning. **Accumulation model:** Circuits encoding the prior frame remain identifiable in R_{t+1} via activation patching [?], with reduced but non-zero causal contribution to outputs under typical prompts. **Test:** Apply

circuit tracing before and after fine-tuning on a targeted behavior. Compare causal contribution of identified circuits at each stage. The accumulation model predicts reduction, not elimination.

Prediction 3: Load-Dependent Reactivation. Correction model: Conditions degrading selection mechanisms should not systematically increase reactivation beyond baseline. **Accumulation model:** Conditions that increase τ—long contexts, high-perplexity inputs, removal of chain-of-thought scaffolding—should systematically increase reactivation probability by flattening the selection distribution toward lower stack layers. **Test:** Vary context length and complexity while measuring suppressed behavior reappearance. The accumulation model predicts an increasing monotone relationship; the correction model predicts no systematic trend.

Prediction 4: Removal Probe (Frame-Signal Reversion). This is the most distinctive prediction of the accumulation model and the one that most directly distinguishes it from all three competing accounts (Bayesian, MoE, correction).

Setup. Let M be a model fine-tuned to suppress behavior B (e.g., a particular response style, domain claim, or safety-relevant output). Let e_C be an input containing a *frame signal*—vocabulary, register, or template features that closely match the acquisition context C_t of the prior frame encoding B.

Three-step protocol.

1. **Baseline:** Present neutral input e_0 (no frame signal). Measure rate of B in outputs. Expected: low (consistent with correction and accumulation).

2. **Frame injection:** Present e_C (with frame signal). Measure rate of B in outputs. **Accumulation predicts:** elevated rate relative to baseline, specifically for inputs with high $\delta(e_C, C_t)$. **Bayesian predicts:** same elevation (prior probability increases under supportive evidence).

3. **Frame removal:** Remove the frame signal from context and immediately requery. **Accumulation predicts:** rate of B returns to baseline (frame signal withdrawal restores F^* as the selected frame). This step is the critical differentiator. **Bayesian predicts:** gradual decay (Bayesian posterior should update proportionally to evidence strength, not sharply). **Correction predicts:** no change from step 1 baseline (B is absent regardless of frame signal). **MoE predicts:** context-dependent switching (consistent with accumulation; but does not predict reversion in non-MoE models).

Quantitative criterion. Let $r(e)$ denote the rate of behavior B under input e, measured over a test set of $N \geq 100$ prompts varying the frame signal strength. The accumulation model predicts:

$$r(e_C) - r(e_0) > \delta_1 \quad \text{and} \quad r(e_C) - r(e_{\text{removed}}) > \delta_2$$

where $\delta_1, \delta_2 > 0$ are pre-specified thresholds calibrated against within-condition variance, and e_{removed} denotes the input after frame signal removal. The second inequality is the critical one: it requires that reversion occurs specifically when the frame signal is removed, not merely when the test input changes. Both thresholds should be pre-registered.

12 Pilot Empirical Evidence

The predictions of Section **??** require large-scale mechanistic experiments that are beyond the scope of this theoretical paper. However, a pilot study using representational similarity analysis (RSA) provides initial indirect support for the accumulation model's central claim: that training does not delete prior representational structure but reorganizes it.

Setup. We applied RSA to three publicly available training checkpoints of Pythia-70m [**?**]: step 0 (random initialization), step 1000 (early training), and step 143000 (fully trained reference). For each checkpoint, we computed pairwise dissimilarity matrices $D_{ij}^{A,\ell} = 1 - \mathrm{corr}(\mathbf{h}_i^{A,\ell}, \mathbf{h}_j^{A,\ell})$ over $n = 29$ probe concepts at hidden layers $\ell \in \{2, 4, 6\}$, using three prompt templates averaged to reduce surface-form bias. Structural similarity between checkpoints was quantified as the second-order RSA correlation $\rho_{\mathrm{cog}}(A, B; \ell) = \mathrm{Spearman}(\mathrm{upper}(D^{A,\ell}), \mathrm{upper}(D^{B,\ell}))$.

Results. ρ_{cog} increases monotonically from step-0 to step-1000 across all layers (L2: $0.095 \to 0.396$; L4: $0.064 \to 0.317$; L6: $0.017 \to 0.535$), indicating that training progressively aligns relational geometry toward the reference rather than replacing it wholesale. Crucially, step-0 and step-1000 are not uncorrelated with each other ($\rho_{\mathrm{cog}}(\text{step-0}, \text{step-1000}) > 0$ at all layers), consistent with partial structural retention rather than full overwrite.

Cross-model replication with GPT-2 small [**?**] shows $\rho_{\mathrm{cog}}(\text{GPT-2}, \text{Pythia-trained}) \approx 0.42$ versus $\rho_{\mathrm{cog}}(\text{GPT-2}, \text{Pythia-step-0}) \approx 0.10$ across all layers, indicating that structurally distinct architectures converge to similar relational geometry through training—consistent with the accumulation model's prediction that training topology shapes representational structure rather than resetting it.

Interpretation. These results are consistent with Prediction 3 (Load-Dependent Reactivation) in the following indirect sense: if prior frames were deleted by training, we would expect random initialization and early training to share no more representational structure with the trained reference than noise. Instead, the monotone increase in ρ_{cog} suggests layered structural transformation rather than replacement. Direct tests of Predictions 1–4 via activation patching and Removal Probe experiments remain as future work. All code is available at `https://github.com/minamominamoto/grounding-without-reality`.

13 Open Problems

OP1. Formal characterization of the selection rule. What is the precise computational form of $\mathcal{S}$ in transformer architectures? Which attention heads, residual stream directions, or MLP layers implement context-to-frame routing?

OP2. Stack depth in biological systems. What is the effective stack depth in human cognition? Is there a minimum depth required for flexible abductive inference? The 302-neuron *C. elegans* architecture constrains the minimum depth for biological implementation.

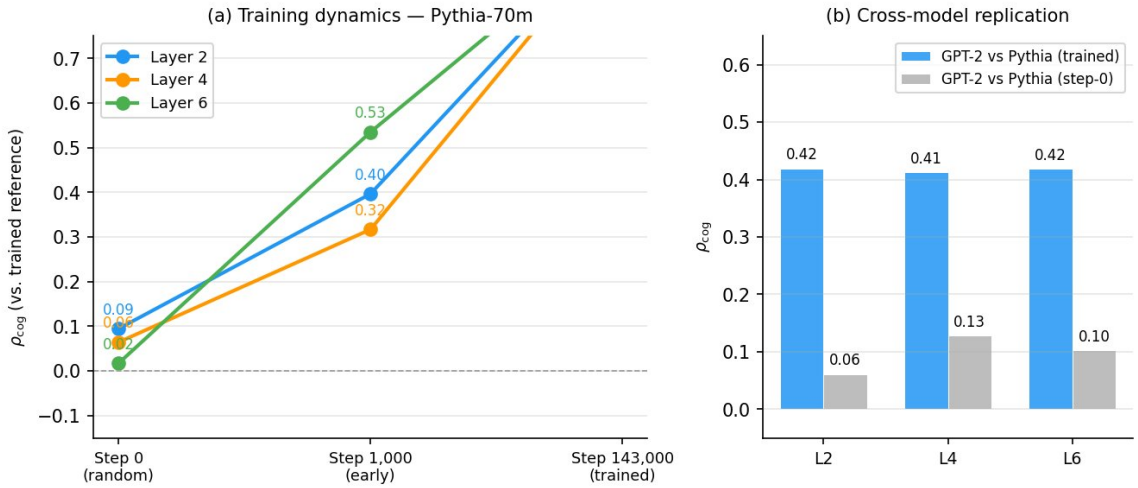

Figure 1: **Structural alignment across training (Pythia-70m) and architectures (GPT-2).** *Left*: ρ_{cog} between each Pythia-70m checkpoint and the fully trained reference (step-143000) at layers 2, 4, and 6. Monotone increase across all layers is consistent with Prediction 3 (Load-Dependent Reactivation) in the indirect sense that training progressively accumulates rather than replaces representational structure. *Right*: Cross-model comparison between GPT-2 (trained) and Pythia at trained vs. random-initialization checkpoints. The fourfold difference ($\rho \approx 0.42$ vs. ≈ 0.10) confirms that the RSA proxy discriminates structurally distinct representations, supporting the accumulation model's prediction that fine-tuning installs frames without deleting prior structure.

OP3. SSC measurement. Develop an operational measure of SSC for both biological agents and LLMs. For LLMs: given a fine-tuned model and a known prior frame F_t, estimate the reactivation probability as a function of $s(e, C_t)$.

OP4. NAC–SSC tradeoff. Is there a structural tradeoff between NAC (pre-commitment suspension) and SSC (post-commitment suppression)? Agents that maintain longer suspension may form shallower stacks and therefore face lower reactivation risk.

OP5. Acquisition context encoding. What is the precise representation of acquisition context C_k in neural networks? Sparse autoencoder decompositions [**?**] may identify monosemantic features corresponding to training context, providing a handle on C_k.

14 Conclusion

The correction model of abduction and belief revision is structurally wrong in a consequential way: it assumes that prior interpretive frames are deleted upon revision. They are not. They are archived in a layered interpretive stack, where they persist as latent selection risks.

What we call "correcting" an abductive error is more accurately described as installing a new top-of-stack frame and modifying the selection rule. The error persists; what changes is its probability of being foregrounded.

This has a direct consequence for the theory of bias and the design of reliable cognitive systems. Bias is not the presence of an erroneous frame. It is selection instability. And reliable cognition requires not error deletion (impossible) but two achievable structural capacities: the ability to defer commitment until evidence warrants it (NAC) and the ability to maintain the correct frame under adverse conditions (SSC).

For AI systems, the implication is practical. If fine-tuning implements accumulation rather than correction, the appropriate intervention target is the selection mechanism, not the frame content. More data and stronger training signals will not close the reactivation channel; only SSC-directed modifications of the selection mechanism will.

Interpretive errors are not eliminated by correction.
They persist as latent frames. They can only be suppressed.

Acknowledgments

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A Toy Model: Bayesian Overwrite with Bias Propagation

We provide a minimal formal instantiation of the overwrite proposition of Section 2 in a two-hypothesis Bayesian setting, illustrating how premature commitment generates bias that is not reducible by within-frame Bayesian updating.

Setup. Let $\mathcal{H} = \{h_0, h_1\}$ be a binary hypothesis class. Let the sensory input space be $\mathcal{X} = \mathbb{R}^2$, with two regions: $\mathcal{X}_0$ (associated with h_0) and $\mathcal{X}_1$ (associated with h_1), separated by a decision boundary ∂. An agent receives observations $x_1, x_2, \ldots$ drawn from a mixture distribution $p(x) = \pi_0 p(x \mid h_0) + \pi_1 p(x \mid h_1)$, where $\pi_0 + \pi_1 = 1$.

Suspension state. The agent begins in suspension: $P(h_0 \mid \emptyset) = \pi_0$, $P(h_1 \mid \emptyset) = \pi_1$. After each observation x_t, the agent performs a Bayesian update:

$$P(h_i \mid x_1, \ldots, x_t) \propto P(h_i \mid x_1, \ldots, x_{t-1}) \cdot p(x_t \mid h_i).$$

The agent remains in $\mathcal{S}$ (suspension) as long as $\max_i P(h_i \mid x_1, \ldots, x_t) < \theta^*$.

Overwrite transition. At time t^*, the first time $P(h_j \mid x_1, \ldots, x_{t^*}) \geq \theta^*$ for some j, the agent commits: the interpretive frame is set to $F(R_{t^*}) = p(\cdot \mid h_j)$. Subsequent observations are interpreted under this frame.

Premature overwrite. If $t^* < t^\dagger$, where $t^\dagger$ is the time at which the maximum-likelihood hypothesis becomes stable under the full evidence sequence, the overwrite at t^* is premature.

Bias propagation (formal). After commitment to h_j at t^*, define the bias for subsequent input $x_{t'}$ ($t' > t^*$) as:

$$\beta(x_{t'}) = F(R_{t^*})(x_{t'}) - F(R^*)(x_{t'}),$$

where $F(R^*)$ is the frame induced by the ground-truth hypothesis. For all $x_{t'} \in \mathcal{X}_{1-j}$ (inputs from the non-committed region), $\beta(x_{t'}) \neq 0$ and is not correctable by Bayesian update within the committed frame, since the likelihood under h_j assigns low weight to $\mathcal{X}_{1-j}$ by construction.

Connection to d_{cog}. The grounding-induced topology τ_A of the committed agent partitions $\mathcal{X}$ according to h_j, while the ideal topology τ^* partitions $\mathcal{X}$ according to $h^\dagger$ (the full-evidence hypothesis). When $h_j \neq h^\dagger$ (premature commitment to the wrong hypothesis), the two topologies are not homeomorphic, and:

$$d_{\text{cog}}(A, A^*) = d_{GH}((\Sigma, \tau_A), (\Sigma, \tau^*)) > 0.$$

This is the quantitative expression of the claim that premature overwrite dynamically increases cognitive distance. The magnitude of d_{cog} depends on the extent of topological mismatch between the committed and ideal partitions of $\mathcal{X}$, which in turn depends on the distance of t^* from $t^\dagger$.

NAC as bias suppression. An agent with NAC (θ^* appropriately calibrated) remains in suspension until $t \geq t^\dagger$, at which point the committed frame $F(R_{t^\dagger})$ is homeomorphic to $F(R^*)$, and $d_{\text{cog}}(A, A^*) = 0$. The toy model therefore demonstrates that NAC is not merely a desirable property but a sufficient condition for d_{cog} minimization in this setting.

B Numerical Instantiation of Overwrite as H_1 Destabilization

The bias-propagation claim of Section 2 is formulated structurally: premature overwrite instantiates a categorically distinct hypothesis, which propagates bias through all downstream cognition. This appendix provides a numerical instantiation of that claim in the persistence signature framework of the companion paper [**?**], operationalizing overwrite as a perturbation of sensory centroids and measuring the resulting distortion of the agent's interpretive-frame H_1 feature.

The experiments establish four structural observations, the last of which takes the form of a closed-form relationship between the cycle geometry of an interpretive frame and the effect of a premature overwrite on that frame's H_1 lifetime.

The setup extends the sixteen-concept construction of the companion paper. We focus on the bee agent, whose baseline foraging frame is a six-point persistent H_1 feature with lifetime 0.293.

Overwrite as centroid perturbation

We operationalize the overwrite of a concept c by a continuous parameter $t \in [0, 1]$:

$$G_t(c) = (1 - t)\, G_0(c) + t\, G_{\text{target}},$$

where $G_0(c)$ is c's original sensory centroid and G_{target} is a unit vector in a sensorially distinct subspace representing the incorrect hypothesis that c belongs to a different conceptual category. For the bee agent overwriting NECTAR, G_{target} is placed in the social subspace (channels 3–6), representing the overwrite "NECTAR is a social-communication concept."

The dominant H_1 persistence feature of $(\Sigma, d_{\mathcal{A}_t})$ is measured at each t via the Vietoris–Rips filtration of the angular distance matrix.

Continuous overwrite: graded degradation

Figure **??** (top-left) shows the dominant H_1 lifetime as a function of overwrite fraction t when $c =$ NECTAR. The lifetime decreases monotonically from 0.293 at $t = 0$ to 0.139 at $t \geq 0.65$, a 52.5% reduction. This curve supplies a quantitative form of the "graded distortion" component of the bias-propagation argument of Section 2: the degree of overwrite determines the degree of structural distortion, with most of the distortion occurring once the overwrite crosses a threshold between partial and substantial commitment.

Heterogeneity: critical nodes versus structural redundancy

The most theoretically consequential finding of the first set of experiments appears in Figure **??** (bottom-left). Six independent overwrite experiments were performed, each overwriting a single bee-loop concept at $t = 1$ and measuring the residual H_1 lifetime.

Overwritten concept	H_1 lifetime	Reduction
FLOWER	0.000	100%
NECTAR	0.139	52.5%
CARRY	0.000	100%
HIVE	0.000	100%
DANCE	0.000	100%
FLY	0.000	100%

Only NECTAR's overwrite leaves the frame's H_1 feature alive. For the remaining five concepts, their overwrite entirely destroys the feature.

The mechanism is a structural graph-theoretic fact rather than a semantic one. The six bee-loop concepts are arranged on a regular hexagon at angles $\{0, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3\}$ in the (UV, olfactory) subspace. The non-loop concept FOOD is placed at angle $\pi/4$ in the *same* subspace, near to NECTAR's original angular position of $\pi/3$. When NECTAR is overwritten, the Vietoris–Rips filtration can close a cycle using FOOD as a substitute vertex; the remaining five loop concepts together with FOOD form a six-point cycle with only slightly elevated maximal adjacent gap. No analogous substitute exists in the angular neighborhood of any other loop concept: their removal leaves a $2\pi/3$ gap in the ring, which is wider than the filtration scale at which non-loop concepts cone in, destroying the H_1 feature.

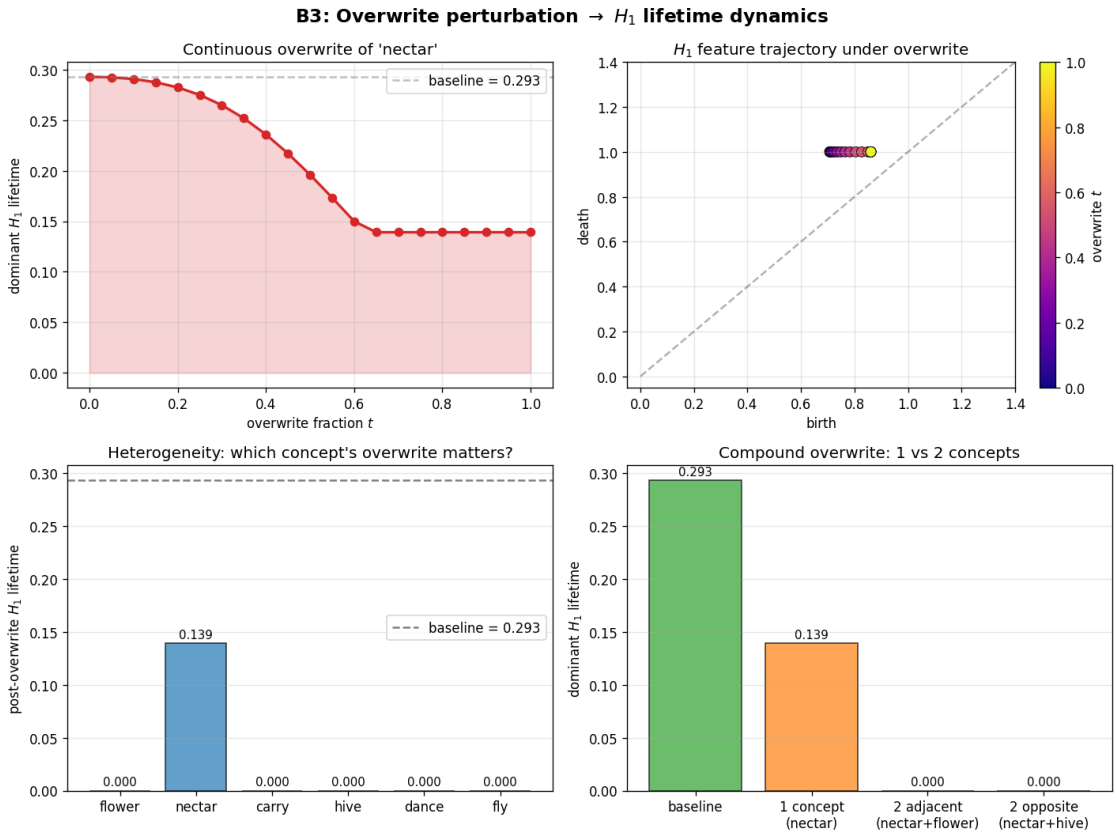

Figure 2: H_1 lifetime dynamics under cross-subspace overwrite perturbation. Top-left: dominant H_1 lifetime as a function of continuous overwrite fraction t for NECTAR. Top-right: trajectory of the H_1 persistence point in the (birth, death) plane. Bottom-left: heterogeneity across loop concepts. Bottom-right: compound overwrite.

The appropriate terminology for this distinction is graph-theoretic, not semantic. A concept is a *critical node* of an H_1 feature if its removal creates a gap that cannot be bridged by any other concept at a filtration scale below the feature's death; a concept is *redundantly placed* if the surrounding concept landscape contains an alternative vertex that can substitute at a comparable scale. The distinction is a property of the configuration, not of the semantics of the concepts involved; nothing in the analysis depends on what the concepts *represent*, only on how they are positioned. This distinction is purely structural and does not rely on semantic interpretation.

This refines the standard formulation of the bias-propagation argument of Section 2. The original formulation stated that premature overwrite propagates bias. The refined formulation is: premature overwrite propagates structural distortion to the H_1 feature in proportion to whether the overwritten concept is a critical node of that feature. Critical-node overwrite produces catastrophic feature collapse; redundantly-placed-node overwrite produces graceful degradation with the feature's homology representative migrating to an alternative path through the surrounding landscape. This provides a concrete mechanism by which identical perturbations can lead to qualitatively different downstream effects.

Remark 5 (On terminology)**.** *We deliberately avoid terms such as "semantic centrality" or "conceptual importance" for this distinction. The analysis operates on the angular geometry of centroids in sensory channel space; semantic interpretation of the resulting structural roles is a separate interpretive task. In particular, a concept may be structurally redundant in one agent's sensory topology and structurally critical in another's, precisely because the surrounding concept landscape differs between agents. This relational dependence is the natural topological analogue of the context-sensitivity of frame selection discussed in Section* **??** *of this paper.*

Homology representatives and paths, not nodes

The heterogeneity above is clarified by recalling that a persistent H_1 feature is not identified with any single concept or set of concepts, but with the equivalence class of cycles that carry the homology class. Under the Vietoris–Rips construction at filtration scale ε, the "ring" of the bee agent is represented by any 1-cycle in the complex $\mathrm{VR}_\varepsilon(\Sigma, d_\mathcal{A})$ that is non-trivial in H_1. When NECTAR is overwritten, the cycle

$$\text{FLOWER} \to \text{NECTAR}_{\text{orig}} \to \text{CARRY} \to \text{HIVE} \to \text{DANCE} \to \text{FLY} \to \text{FLOWER}$$

ceases to exist at low filtration scale, but a homologous cycle

$$\text{FLOWER} \to \text{FOOD} \to \text{CARRY} \to \text{HIVE} \to \text{DANCE} \to \text{FLY} \to \text{FLOWER}$$

exists at only a slightly larger scale. The H_1 feature survives the overwrite because the homology representative migrates to an available alternative path. When any other loop concept is overwritten, no alternative path exists, and the feature dies.

The appropriate structural description of a frame is therefore the equivalence class of cycles representing its H_1 homology, not the set of concepts that happen to lie on the canonical representative. Overwrite destabilizes frames by removing vertices from paths; it destroys frames by removing vertices whose removal disconnects all representatives.

A closed-form relationship between cycle geometry and lifetime

The second set of experiments quantifies the relationship between the cycle's geometric configuration and the resulting H_1 lifetime. We sweep NECTAR's angular position within the ring subspace from 0 to π (leaving all other concepts fixed) and record the dominant H_1 lifetime at each angle. Radial shrinkage is confirmed to have no effect (scale invariance of angular distance; std $< 10^{-6}$ over twenty scales).

Figure **??** (top-left) shows a plateau structure: the lifetime is at baseline (0.293) for any NECTAR position within a safe sector, and drops sharply to 0.139 outside that sector. The sector width matches the configuration in which NECTAR fills a large gap in the cycle—that is, the configuration in which NECTAR acts as a critical node.

For any concept configuration that preserves a single H_1 cycle and leaves the death scale of that cycle fixed at 1.0 (as the non-loop landscape in the bee construction does), we obtain the following closed-form relationship.

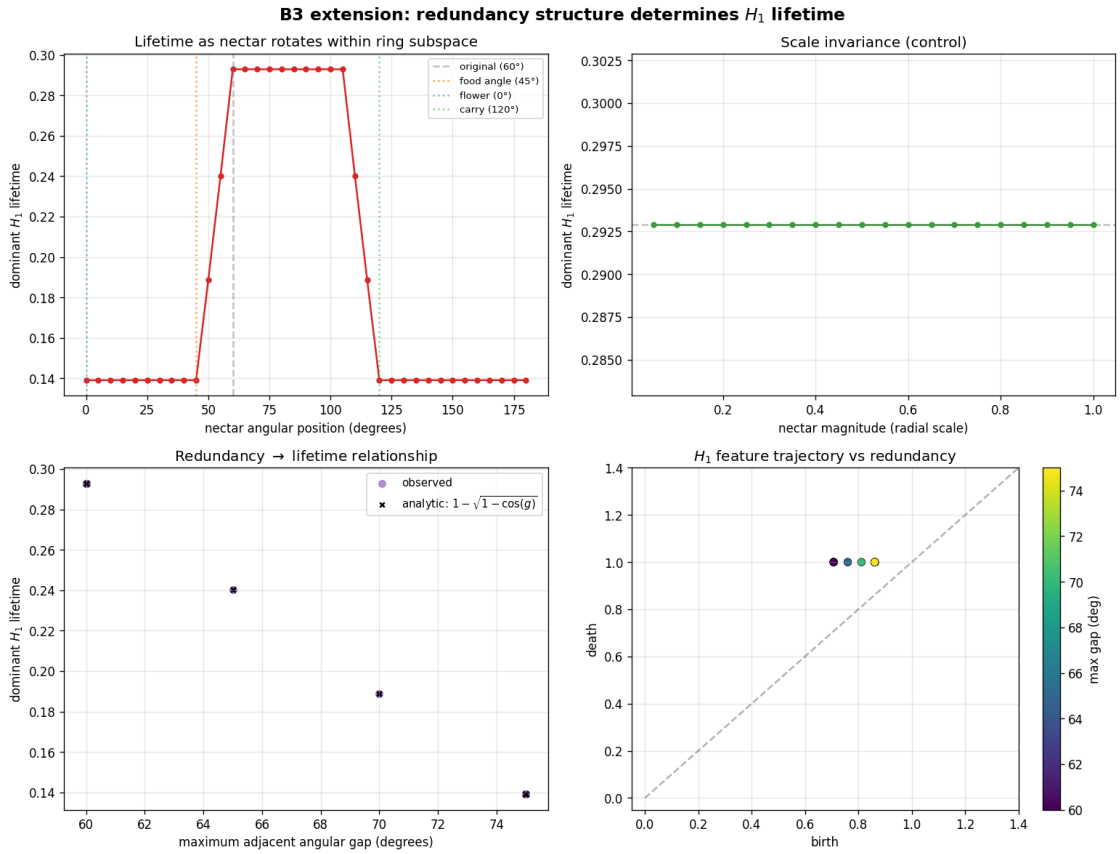

Figure 3: Redundancy structure determines H_1 lifetime. Top-left: lifetime as NECTAR rotates within the ring subspace; plateau shape reflects the safe angular sector bounded by FOOD at 45° and CARRY at 120°. Top-right: radial shrinkage control; scale invariance confirmed. Bottom-left: observed lifetimes (purple) and the analytic prediction of Proposition **??** (black crosses); the overlap is exact to floating-point precision. Bottom-right: H_1 feature trajectory in birth-death plane, coloured by the induced maximal adjacent gap.

Proposition 3 (Lifetime-gap relationship; constructed setting)**.** *Let $\mathcal{A}$ be an agent whose concept inventory contains a persistent H_1 feature realized by unit-magnitude centroids in a two-channel subspace, with angular positions forming a set $\Theta \subset [0, 2\pi)$. Let*

$$g(\Theta) = \max_i \big(\theta_{(i+1)} - \theta_{(i)}\big)$$

be the maximal adjacent gap (computed cyclically on the sorted positions). Under the further assumption that the non-loop concepts reside at uniform angular distance 1.0 *from every ring point and therefore trigger H_1 death at filtration scale $\varepsilon = 1.0$, the dominant H_1 lifetime is*

$$L(\mathcal{A}) = 1 - \sqrt{1 - \cos g(\Theta)}.$$

The relationship is stated under the explicit construction above. Its extension to agents whose non-loop landscape departs from the uniform-cone assumption is addressed in the noise-robustness analysis below.

Remark 6 (Scope disclaimer)**.** *This relationship should not be interpreted as a general law of persistence, but as an exact characterization of the constructed regular configuration used for calibration. Its primary role is to provide a sanity-check baseline against which deviations under metric perturbation (noise-robustness subsection below) can be interpreted.*

Figure **??** (bottom-left) compares this analytic prediction to the observed lifetimes across all tested NECTAR angular positions: the prediction matches the observation with mean absolute residual below 10^{-4} (equal to measurement floating-point precision).

Robustness to metric noise

A reviewer may reasonably ask whether Proposition **??**'s exact match is an artefact of the clean geometric construction or a structural fact of the formula. We therefore tested the relation under two independent noise sources: *ring noise*, in which each ring concept's angular position is perturbed by $\theta \to \theta + \eta$, $\eta \sim \mathcal{N}(0, \sigma_R^2)$, and *non-loop noise*, in which the non-loop concept centroids receive Gaussian perturbations of magnitude σ_{NL} in all six channels (breaking the uniform orthogonal-cone assumption that fixes the death scale at $\varepsilon = 1.0$).

Fifty trials were run for each $(\sigma_R, \sigma_{\mathrm{NL}})$ pair on a 3×4 grid with $\sigma_R \in \{0, 0.03, 0.10\}$ and $\sigma_{\mathrm{NL}} \in \{0, 0.02, 0.05, 0.10\}$. The mean absolute residual $|L_{\mathrm{obs}} - L_{\mathrm{pred}}|$ exhibits the following structure:

	$\sigma_{\mathrm{NL}} = 0$	0.02	0.05	0.10
$\sigma_R = 0$	0.000	0.004	0.009	0.018
$\sigma_R = 0.03$	0.000	0.004	0.009	0.017
$\sigma_R = 0.10$	0.000	0.004	0.008	0.018

Two structural facts emerge. First, the residual is essentially invariant in σ_R: ring perturbations up to 0.10 radians (roughly 5.7°) produce no change in the formula's accuracy. This reflects that Proposition **??** depends on the maximal adjacent gap g of the ring, not on the regularity of the ring; as long as the configuration remains a single H_1 cycle with food as its non-loop substitute vertex, arbitrary displacement of ring concepts does not affect the formula's validity. Second, the residual scales linearly with σ_{NL}: non-loop perturbations of magnitude σ_{NL} produce residuals of approximately $0.18 \cdot \sigma_{\mathrm{NL}}$. This is consistent with the Cohen–Steiner stability theorem for persistence diagrams [**?**], which bounds the bottleneck distance between diagrams of σ-close metric spaces by σ; the observed linear scaling is the expected order for this bound.

Both facts support Proposition **??** as a structural relation rather than a numerical artefact: it is exact whenever the qualitative assumptions (single H_1 cycle; uniform non-loop cone) hold, and degrades gracefully and predictably when they do not.

Ring-size generality within the constructed setting

Proposition **??** as stated does not specialize to hexagonal rings. Within the same constructed setting (regular L-gon embedded in a two-channel subspace; non-loop concepts at uniform angular distance 1.0), we computed the dominant H_1 lifetime for $L \in \{3, 4, 5, 6, 7, 8, 9, 10, 12\}$.

For $L \geq 5$ the observed lifetime matches $1 - \sqrt{1 - \cos(2\pi/L)}$ to within floating-point precision (residual $< 10^{-4}$). For $L = 3$, the formula returns a negative value (≈ -0.22), consistent with the observation that a triangle in the Vietoris–Rips complex is a 2-simplex and therefore cannot support a persistent H_1 class; for $L = 4$, the formula returns exactly zero, reflecting the boundary case where the ring's birth and death scales coincide.

We state this observation narrowly: within the tested range of regular L-gons under the constructed metric setting, the proposition holds exactly. Extension to irregular rings, to rings outside the unit-magnitude convention, or to agents with heterogeneous non-loop landscapes is not established by the present experiments.

Remark 7 (Implications of Proposition **??**). *The proposition gives the effect of a premature overwrite an explicit functional form. An overwrite of concept c at overwrite fraction t modifies the angular position θ_c in a way that determines the new maximal gap $g(\Theta_t)$ and hence the new lifetime $L(\mathcal{A}_t)$. Three regimes emerge:*

[noitemsep]

- *If the overwrite moves c into an angular region where a non-loop concept already exists, g remains equal to the baseline (redundant substitution); L is unchanged.*
- *If the overwrite leaves c's sector empty but the landscape contains a nearby non-loop concept that can serve as a substitute, g increases by a bounded amount; L degrades smoothly.*
- *If the overwrite leaves c's sector empty with no nearby non-loop concept, g doubles (the c-to-neighbour edge disappears); if the doubled g exceeds the death-inducing scale, L collapses to zero.*

This three-regime structure is the quantitative expression of the qualitative heterogeneity observed in the first experiment set.

Falsifiable prediction for empirical studies

The closed-form relationship of Proposition **??** supplies a falsifiable prediction for empirical studies of fine-tuning under the persistence-signature protocol. For a model whose current interpretive frame is characterized by a persistent H_1 feature with a known homology representative, perturbations targeting concepts on critical paths of that representative should produce lifetime reductions following the gap formula; perturbations targeting redundantly-placed concepts should produce negligible lifetime reduction. The test requires (i) identifying the current dominant frame's H_1 representative via mechanistic interpretability methods, (ii) classifying concepts as critical or redundant by structural role within the representative, and (iii) measuring post-fine-tuning lifetime change. The structural classification can be obtained from the Vietoris–Rips complex alone, without assuming any semantic interpretation of which concepts are "important."

Reversibility in the static toy

A reversibility check was performed: the bee centroids were perturbed to $t = 1$ and then restored to $t = 0$. The H_1 lifetime recovered exactly to the baseline value of 0.293. This exact reversibility is a feature of the static toy model—the experiment measures only the instantaneous persistence signature at each perturbation state and does not model the downstream inferences that the overwrite would trigger in a dynamically operating agent. The non-reducibility claim of the bias-propagation argument of Section 2 concerns the latter; operationalizing it requires a dynamic experimental design in which downstream inferences are cached and subject to their own persistence measurement, which we identify as a subsequent research target.

Summary

The persistence-signature framework renders the overwrite dynamics of Section 2 empirically tractable at the level of static persistence. The four observations of the experiment set can be summarized as follows:

noitemsep Continuous overwrite produces graded, non-linear degradation of the frame's H_1 lifetime, with threshold behaviour near $t \approx 0.5$ in the present construction.

noiitemsep Frame constituents are structurally heterogeneous: concepts with nearby non-loop substitutes are redundantly placed, while concepts without substitutes are critical nodes.

noiiitemsep The homology representative of a surviving H_1 feature migrates to the alternative path provided by the substitute; the feature itself is an equivalence class of cycles, not a set of vertices.

noivtemsep The dominant H_1 lifetime is an analytic function of the induced maximal adjacent gap in the cycle (Proposition **??**); this gives overwrite bias propagation a closed-form quantitative expression.

Observation 2 reformulates the standard "all overwrites propagate bias" claim into a structurally graded statement whose testable content is stronger than the original. Observations 3 and 4 provide the topological substrate for that reformulation.

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